Unit 4 Review

Solving Problems with Rational Numbers

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers each part of the assessment, with more questions than the assessment has. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Adding and Subtracting in Scientific Notation

Lesson 1 · 6 questions

1. $(4.1 \times 10^{6}) + (2.3 \times 10^{6})$

2. $(7.6 \times 10^{-4}) - (2.9 \times 10^{-4})$

3. $(8.2 \times 10^{5}) + (4.9 \times 10^{5})$

4. $(5.4 \times 10^{8}) + (7.1 \times 10^{7})$

5. $(6.3 \times 10^{-2}) - (8 \times 10^{-4})$

6. Lena writes $(3.2\times10^4)+(5.5\times10^6)=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

PART B. Multiplying and Dividing in Scientific Notation

Lessons 2–3 · 6 questions

7. $(4\times10^3)(2\times10^6)$

8. $\frac{7.2\times10^8}{1.8\times10^3}$

9. $(5\times10^6)(8\times10^{-2})$

10. $\frac{2.4\times10^5}{8\times10^9}$

11. $\frac{4\times10^7}{4\times10^{-3}}$

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4×10^{2} seconds?

PART C. Real-World Problems in Scientific Notation

Lesson 4 · 4 questions

13. A rectangular panel measures 3.2×10^{2} mm by 1.45×10^{2} mm. Find its area, in scientific notation. Do not round.

14. A train travels 4.8×10^{2} km in 6 hours. Find its average speed, in scientific notation. Do not round.

15. Two lengths are measured: 5.4×10^{3} mm and 2.8×10^{2} mm. Find their sum, in scientific notation. Do not round.

16. Two masses are measured: 7.6×10^{-2} g and 3.42×10^{-3} g. Find their difference, in scientific notation. Do not round.

PART D. Evaluating Expressions with Radicals

Lesson 5 · 7 questions

17. $\sqrt{169}$

18. $\sqrt[3]{-8}$

19. $\sqrt{7^2 - 24}$

20. $\sqrt[3]{\frac{1}{27}}$

21. $\sqrt{100} \cdot 4^{-2}$

22. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2 + 7}$

23. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

PART E. The Order of Operations

Lesson 6 · 7 questions

24. $24 - 3 \cdot 5 + 1$

25. $48 \div 6 \cdot 2$

26. $4 + 3(8 - 5)^2$

27. $\frac{6^2 + 9}{3^2 - 4}$

28. $-6^2 + (-6)^2$

29. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

30. Which expression is equivalent in value to $2+4\cdot5^2$?

- ○ A. 900
- ○ B. 102
- ○ C. 42
- ○ D. 150

PART F. Real-World Problems and the Order of Operations

Lessons 7–8 · 5 questions

31. Evaluate $5x^2$ when $x=-3$.

32. Kinetic energy is $K=\frac{1}{2}mv^2$. Find K when $m=400$ kg and $v=7$ m/s.

33. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

34. Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery.

a. What is the area of the patio?

b. Write an expression for the total cost.

c. Find the total cost.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

35. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ○ A. $18^2(3.50 + 140)$
- ○ B. $18(3.50) + 140$
- ○ C. $18^2 + 3.50 + 140$
- ○ D. $18^2(3.50) + 140$

GRADE 7 ACCELERATED · UNIT 4

Unit 4 Review

Solving Problems with Rational Numbers **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers each part of the assessment, with more questions than the assessment has. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Adding and Subtracting in Scientific Notation

Lesson 1 · MA.8.NSO.1.5 · 6 questions

1. $(4.1\times10^{6})+(2.3\times10^{6})$

6.4×10^{6}

Same power: add the coefficients, copy the power down.

2. $(7.6\times10^{-4})-(2.9\times10^{-4})$

4.7×10^{-4}

3. $(8.2\times10^{5})+(4.9\times10^{5})$

1.31×10^{6}

13.1×10^{5} *is the value and not the finished form.*

4. $(5.4\times10^{8})+(7.1\times10^{7})$

6.11×10^{8}

$7.1\times10^{7}=0.71\times10^{8}$.

5. $(6.3\times10^{-2})-(8\times10^{-4})$

6.22×10^{-2}

$8\times10^{-4}=0.08\times10^{-2}$ *; the exponents are two apart, the widest gap allowed.*

6. Lena writes $(3.2\times10^{4})+(5.5\times10^{6})=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

She added the coefficients without first writing both over the same power of ten, and she added the exponents. The sum is 5.532×10^{6} **.**

$3.2\times10^{4}=0.032\times10^{6}$ *, and* $5.5+0.032=5.532$.

PART B. Multiplying and Dividing in Scientific Notation

Lessons 2–3 · MA.8.NSO.1.5 · 6 questions

7. $(4\times10^{3})(2\times10^{6})$

8×10^{9}

Multiply the coefficients, add the exponents.

8. $\frac{7.2\times10^{8}}{1.8\times10^{3}}$

4×10^{5}

Divide the coefficients, subtract the exponents, top minus bottom.

9. $(5\times10^{6})(8\times10^{-2})$

4×10^{5}

40×10^{4} *is not finished.*

10. $\frac{2.4\times10^{5}}{8\times10^{9}}$

3×10^{-5}

0.3×10^{-4} *is not finished either.*

11. $\frac{4\times10^{7}}{4\times10^{-3}}$

1×10^{10}

$7-(-3)=10$ *. Subtracting a negative goes up.*

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4×10^{2} seconds?

2.6×10^{17} ***photons***

$6.5\cdot4=26$ *, so* 26×10^{16} *before it is finished.*

PART C. Real-World Problems in Scientific Notation

Lesson 4 · MA.8.NSO.1.6 · 4 questions

13. A rectangular panel measures 3.2×10^{2} mm by 1.45×10^{2} mm. Find its area, in scientific notation. Do not round.

4.64×10^{4} ***square millimeters***

$3.2\cdot1.45=4.64$ *, and* $10^{2}\cdot10^{2}=10^{4}$ *.*

14. A train travels 4.8×10^{2} km in 6 hours. Find its average speed, in scientific notation. Do not round.

8×10^{1} ***km/h***

$\frac{4.8\times10^{2}}{6}=0.8\times10^{2}$ *, which is not finished.*

15. Two lengths are measured: 5.4×10^{3} mm and 2.8×10^{2} mm. Find their sum, in scientific notation. Do not round.

5.68×10^{3} ***mm***

$2.8\times10^{2}=0.28\times10^{3}$ *, and* $5.4+0.28=5.68$.

16. Two masses are measured: 7.6×10^{-2} g and 3.42×10^{-3} g. Find their difference, in scientific notation. Do not round.

7.258×10^{-2} ***g***

$3.42\times10^{-3}=0.342\times10^{-2}$ *, and* $7.6-0.342=7.258$.

PART D. Evaluating Expressions with Radicals

Lesson 5 · MA.8.NSO.1.7 · 7 questions

17. $\sqrt{169}$

13

18. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

19. $\sqrt{7^{2}-24}$

5

Finish under the bar: 49 − 24 = 25.

20. $\sqrt[3]{\frac{1}{27}}$

$\frac{1}{3}$

Top and bottom each get the root.

21. $\sqrt{100}\cdot4^{-2}$

$\frac{5}{8}$

$10\cdot\frac{1}{16}$.

22. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2+7}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

23. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

No. A cube root asks for the number used three times, and $(-4)(-4)(-4) = -64$ ***, so the value is −4. It is a square root of a negative number that is not a real number.***

This is the first misconception the state guide names for this benchmark.

PART E. The Order of Operations

Lesson 6 · MA.8.NSO.1.7 · 7 questions

24. $24 - 3 \cdot 5 + 1$

10

Multiply first, then left to right.

25. $48 \div 6 \cdot 2$

16

One level, left to right: 8 · 2. Not 4.

26. $4 + 3(8-5)^2$

31

$4 + 3(9)$.

27. $\frac{6^2+9}{3^2-4}$

9

45 over 5.

28. $-6^2 + (-6)^2$

0

The exponent attaches to its own base; the parentheses decide what that base is.

29. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

9

$20 \cdot \frac{9}{20}$.

30. Which expression is equivalent in value to $2+4\cdot5^2$?

- ○ A. 900
- ● **B.** 102
- ○ C. 42
- ○ D. 150

B

PART F. Real-World Problems and the Order of Operations *Lessons 7–8 · MA.8.NSO.1.7 · 5 questions*

31. Evaluate $5x^2$ when $x=-3$.

45

$5(-3)^2=5(9)$.

32. Kinetic energy is $K=\frac{1}{2}mv^2$. Find K when $m=400$ kg and $v=7$ m/s.

9,800 joules

$200(49)$ *. The unit is part of the answer.*

33. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

78 dollars and 364 dollars

$30+12(4)=78$ *and* $30(12)+4=364$ *. The same three numbers and the same words; what decides is which amount is repeated.*

34. SSDD. Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery. Same surface, four different questions: a measurement, an expression, an evaluation, and a size read back out of a total. Only part b has no number for an answer, and only part d runs the structure backwards.

a. What is the area of the patio?

225 square feet

b. Write an expression for the total cost.

$15^2(2.20)+60$

The rate multiplies the area; the flat fee does not.

c. Find the total cost.

555 dollars

$225(2.20)=495$ *, then* $+60$.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

20 feet

$940-60=880$ *, and* $880\div2.20=400$ *, whose square root is 20.*

35. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ○ A. $18^2(3.50 + 140)$
- ○ B. $18(3.50) + 140$
- ○ C. $18^2 + 3.50 + 140$
- ● **D.** $18^2(3.50) + 140$

D

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ○ C. 2.914×10^5
- ○ D. 4.3×10^8

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☐ A. $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☐ C. $(6 \times 10^3)(5 \times 10^2)$
- ☐ D. $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☐ F. $\frac{9 \times 10^9}{3 \times 10^3}$

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1} . What was the number? Write your answer in scientific notation. *1 pt*

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^{2} mm by 1.35×10^{2} mm. Find its area. *1 pt*

b. A vehicle travels 5.6×10^{2} km in 8 hours. Find its average speed. *1 pt*

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^{3} mm and 4.1×10^{2} mm. Find their sum. *1 pt*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

b. $\frac{5^2 + 7}{2^3}$ *1 pt*

c. $36 \div 4 \cdot 3$ *1 pt*

d. $-4^2 + (-4)^2$ *1 pt*

10. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

b. Find the total cost. *1 pt*

12. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ○ B. 47
- ○ C. 63
- ○ D. 289

13. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☐ B. $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☐ D. $\left(\frac{1}{6}\right)^{-1}$
- ☐ E. $\sqrt{5^2 + 11}$
- ☐ F. $\frac{4^2 + 8}{4}$

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points · two class periods **ANSWER KEY**

Name
Date
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This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points MA.8.NSO.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

5.7×10^5

Same power: add the coefficients.

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

2.5×10^{-6}

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

1.43×10^9

14.3×10^8 *is the value; 14.3 is not less than 10, so it is not the answer.*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

4.57×10^4

$3.7 \times 10^3 = 0.37 \times 10^4$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

6×10^{10}

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

8×10^5

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

2×10^5

20×10^4 *is not finished.*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3×10^{-4}

0.3×10^{-3} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ● **C.** 2.914×10^5
- ○ D. 4.3×10^8

C

Scoring: 1 point.

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

A, C, D and F

C gives 30×10^5 *, which is* 3×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 6.123×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$2.3 \times 10^4 = 0.023 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1}. What was the number? Write your answer in scientific notation. *1 pt*

5×10^{-5}

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed.

$\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.5×10^{-4} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation

4 points MA.8.NSO.1.6

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

3.24×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 5.6×10^2 km in 8 hours. Find its average speed. *1 pt*

7×10^1 ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^3 mm and 4.1×10^2 mm. Find their sum. *1 pt*

6.66×10^3 ***mm***

Match the powers first: $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

8.828×10^{-2} ***g***

Match the powers first: $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

MA.8.NSO.1.7

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

−3

The power first, then the product, then the subtraction.

b. $\dfrac{5^2 + 7}{2^3}$ *1 pt*

4

32 over 8; the bar groups the whole numerator.

c. $36 \div 4 \cdot 3$ *1 pt*

27

One level, left to right: 9 · 3. Not 3.

d. $-4^2 + (-4)^2$ *1 pt*

0

−16 + 16. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

7

Finish under the bar: 36 + 13 = 49.

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5 + \frac{1}{4}$.

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9} + \frac{2}{3}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$14^2(2.40) + 85$

The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

555.40 dollars

196(2.40) = 470.40, then + 85.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

B

Scoring: 1 point.

13. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☒ **B.** $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☒ **D.** $\left(\frac{1}{6}\right)^{-1}$
- ☒ **E.** $\sqrt{5^2 + 11}$
- ☒ **F.** $\frac{4^2 + 8}{4}$

B, D, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken: $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.

Scoring: 1 point.

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

676 square inches

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.NSO.1.5	1, 2, 3, 4, 5, 6	12	
MA.8.NSO.1.6	7, 8	4	
MA.8.NSO.1.7	9, 10, 11, 12, 13, 14, 15	13	
Total		**29**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is

arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

5.7×10^5

Same power: add the coefficients.

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

2.5×10^{-6}

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

1.43×10^9

14.3×10^8 *is the value; 14.3 is not less than 10, so it is not the answer.*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

4.57×10^4

$3.7 \times 10^3 = 0.37 \times 10^4$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

6×10^{10}

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

8×10^5

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

2×10^5

20×10^4 *is not finished.*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3×10^{-4}

0.3×10^{-3} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ● **C.** 2.914×10^5
- ○ D. 4.3×10^8

C

Scoring: 1 point.

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

A, C, D and F

C gives 30×10^5 *, which is* 3×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 6.123×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$2.3 \times 10^4 = 0.023 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1}. What was the number? Write your answer in scientific notation. *1 pt*

5×10^{-5}

$\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.5×10^{-4} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

3.24×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 5.6×10^2 km in 8 hours. Find its average speed. *1 pt*

7×10^1 ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^3 mm and 4.1×10^2 mm. Find their sum. *1 pt*

6.66×10^3 ***mm***

Match the powers first: $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

8.828×10^{-2} ***g***

Match the powers first: $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

−3

The power first, then the product, then the subtraction.

b. $\dfrac{5^2 + 7}{2^3}$ *1 pt*

4

32 over 8; the bar groups the whole numerator.

c. $36 \div 4 \cdot 3$ *1 pt*

27

One level, left to right: 9 · 3. Not 3.

d. $-4^2 + (-4)^2$ *1 pt*

0

−16 + 16. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

7

Finish under the bar: 36 + 13 = 49.

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5 + \frac{1}{4}$.

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9} + \frac{2}{3}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$14^2(2.40) + 85$

The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

555.40 dollars

196(2.40) = 470.40, then + 85.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

B

Scoring: 1 point.

13. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☒ **B.** $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☒ **D.** $\left(\frac{1}{6}\right)^{-1}$
- ☒ **E.** $\sqrt{5^2 + 11}$
- ☒ **F.** $\frac{4^2 + 8}{4}$

B, D, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken: $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.

Scoring: 1 point.

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

676 square inches

The side of the first is $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^{6}) + (2.4 \times 10^{6})$ *1 pt*

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

c. $(6.4 \times 10^{7}) + (8.2 \times 10^{7})$ *1 pt*

d. $(5.3 \times 10^{5}) + (2.6 \times 10^{4})$ *1 pt*

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ○ B. 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

- ☐ A. $(2.5 \times 10^6) + (1.5 \times 10^6)$
- ☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
- ☐ C. $(8 \times 10^3)(5 \times 10^2)$
- ☐ D. $(4.3 \times 10^6) - (3 \times 10^5)$
- ☐ E. $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
- ☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2} . What was the number? Write your answer in scientific notation. *1 pt*

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

b. $\frac{4^2 + 9}{5}$ *1 pt*

c. $45 \div 9 \cdot 3$ *1 pt*

d. $-3^2 - (-3)^2$ *1 pt*

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

b. Find the total cost. *1 pt*

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ○ D. 39

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☐ A. $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☐ C. $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☐ E. $\frac{5^2+7}{4}$
- ☐ F. $\sqrt{6^2+28}$

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points MA.8.NSO.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^{6}) + (2.4 \times 10^{6})$ *1 pt*

6.7×10^{6}

Same power: add the coefficients.

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

5.5×10^{-5}

c. $(6.4 \times 10^{7}) + (8.2 \times 10^{7})$ *1 pt*

1.46×10^{8}

14.6×10^{7} *is the value; 14.6 is not less than 10, so it is not the answer.*

d. $(5.3 \times 10^{5}) + (2.6 \times 10^{4})$ *1 pt*

5.56×10^{5}

$2.6 \times 10^{4} = 0.26 \times 10^{5}$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

8×10^8

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

4×10^5

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

3×10^3

30×10^2 *is not finished.*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3×10^{-5}

0.3×10^{-4} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ● **B.** 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

B

Scoring: 1 point.

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

- ☒ **A.** $(2.5 \times 10^6) + (1.5 \times 10^6)$
- ☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
- ☒ **C.** $(8 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(4.3 \times 10^6) - (3 \times 10^5)$
- ☒ **E.** $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
- ☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

A, C, D and E

C gives 40×10^5 *, which is* 4×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 5.234×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$3.4 \times 10^4 = 0.034 \times 10^6$ *.*

Scoring: 1 point.

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

6×10^{-7}

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed.

$\frac{3 \times 10^{-2}}{5 \times 10^4} = 0.6 \times 10^{-6} = 6 \times 10^{-7}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.6×10^{-6} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation

4 points MA.8.NSO.1.6

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

4.375×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 3.5 · 1.25 = 4.375, and $10^2 \cdot 10^2 = 10^4$ *.*

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

9×10^1 ***km/h***

$\frac{7.2 \times 10^2}{8} = 0.9 \times 10^2$ *, which is not finished: 0.9 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

7.87×10^3 ***mm***

Match the powers first: $5.2 \times 10^2 = 0.52 \times 10^3$ *, and 7.35 + 0.52 = 7.87.*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

8.039×10^{-2} ***g***

Match the powers first: $3.61 \times 10^{-3} = 0.361 \times 10^{-2}$ *, and 8.4 − 0.361 = 8.039.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

MA.8.NSO.1.7

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

8

The power first, then the product, then the subtraction.

b. $\frac{4^2 + 9}{5}$ *1 pt*

5

25 over 5; the bar groups the whole numerator.

c. $45 \div 9 \cdot 3$ *1 pt*

15

One level, left to right: 5 · 3. Not 5.

d. $-3^2 - (-3)^2$ *1 pt*

−18

−9 − 9. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

9

Finish under the bar: 64 + 17 = 81.

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

$-\frac{35}{9}$

$-4 + \frac{1}{9}$.

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

1

$\frac{1}{4} + \frac{3}{4}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$12^2(2.60) + 75$

The rate multiplies the area; the flat fee does not. An expression that charges 75 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

449.40 dollars

144(2.60) = 374.40, then + 75.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ● **D. 39**

D

Scoring: 1 point.

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☒ **A.** $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☒ **C.** $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☒ **E.** $\frac{5^2+7}{4}$
- ☒ **F.** $\sqrt{6^2+28}$

A, C, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 36 + 64 is finished before the root is taken: $\sqrt{100} = 10$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '10' alone does not earn it, and neither does 'he added wrong'.

Scoring: 1 point.

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

900 square inches

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is $\sqrt{225} = 15$ *, so the second has side 30 and area* $30^2 = 900$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 450 has not gone through the side.*

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.NSO.1.5	1, 2, 3, 4, 5, 6	12	
MA.8.NSO.1.6	7, 8	4	
MA.8.NSO.1.7	9, 10, 11, 12, 13, 14, 15	13	
Total		**29**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is

arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation *12 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^{6}) + (2.4 \times 10^{6})$ *1 pt*

6.7×10^{6}

Same power: add the coefficients.

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

5.5×10^{-5}

c. $(6.4 \times 10^{7}) + (8.2 \times 10^{7})$ *1 pt*

1.46×10^{8}

14.6×10^{7} *is the value; 14.6 is not less than 10, so it is not the answer.*

d. $(5.3 \times 10^{5}) + (2.6 \times 10^{4})$ *1 pt*

5.56×10^{5}

$2.6 \times 10^{4} = 0.26 \times 10^{5}$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

8×10^8

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

4×10^5

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

3×10^3

30×10^2 *is not finished.*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3×10^{-5}

0.3×10^{-4} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ● **B.** 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

B

Scoring: 1 point.

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

- ☒ **A.** $(2.5 \times 10^6) + (1.5 \times 10^6)$
- ☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
- ☒ **C.** $(8 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(4.3 \times 10^6) - (3 \times 10^5)$
- ☒ **E.** $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
- ☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

A, C, D and E

C gives 40×10^5 *, which is* 4×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 5.234×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$3.4 \times 10^4 = 0.034 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

6×10^{-7}

$\frac{3 \times 10^{-2}}{5 \times 10^4} = 0.6 \times 10^{-6} = 6 \times 10^{-7}$. *Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.6×10^{-6} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

4.375×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 3.5 · 1.25 = 4.375, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

9×10^1 ***km/h***

$\frac{7.2 \times 10^2}{8} = 0.9 \times 10^2$, *which is not finished: 0.9 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

7.87×10^3 ***mm***

Match the powers first: $5.2 \times 10^2 = 0.52 \times 10^3$ *, and 7.35 + 0.52 = 7.87.*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

8.039×10^{-2} ***g***

Match the powers first: $3.61 \times 10^{-3} = 0.361 \times 10^{-2}$ *, and 8.4 − 0.361 = 8.039.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

8

The power first, then the product, then the subtraction.

b. $\frac{4^2 + 9}{5}$ *1 pt*

5

25 over 5; the bar groups the whole numerator.

c. $45 \div 9 \cdot 3$ *1 pt*

15

One level, left to right: 5 · 3. Not 5.

d. $-3^2 - (-3)^2$ *1 pt*

−18

−9 − 9. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

9

Finish under the bar: 64 + 17 = 81.

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

$-\frac{35}{9}$

$-4 + \frac{1}{9}$.

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

1

$\frac{1}{4} + \frac{3}{4}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$12^2(2.60) + 75$

The rate multiplies the area; the flat fee does not. An expression that charges 75 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

449.40 dollars

144(2.60) = 374.40, then + 75.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ● **D. 39**

D

Scoring: 1 point.

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☒ **A.** $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☒ **C.** $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☒ **E.** $\frac{5^2+7}{4}$
- ☒ **F.** $\sqrt{6^2+28}$

A, C, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 36 + 64 is finished before the root is taken: $\sqrt{100} = 10$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '10' alone does not earn it, and neither does 'he added wrong'.

Scoring: 1 point.

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

900 square inches

The side of the first is $\sqrt{225} = 15$ *, so the second has side 30 and area* $30^2 = 900$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 450 has not gone through the side.*

Scoring: 1 point.